

PORTO AP, Portugal

WMO: 085450

Lat: 41.2322N

Lon: 8.6792W

Elev: 229

StdP: 14.57

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.1	38.5	22.3	16.9	47.2	26.2	20.3	47.0	24.9	55.5	22.0	54.7	5.0	100	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.5	86.9	65.6	82.9	64.4	79.3	63.4	68.8	78.6	67.4	75.9	66.3	73.8	8.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	97.3	70.2	64.8	92.5	68.9	64.0	90.0	68.3	33.0	78.5	31.8	76.1	31.0	73.6	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	18.3	16.0	31.6	94.7	2.1	3.9	30.0	97.5	28.8	99.8	27.6	102.0	26.1	104.8
			29.0	72.5	2.5	2.8	27.2	74.5	25.7	76.1	24.3	77.6	22.5	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	50.5	51.3	54.8	56.8	60.5	65.0	67.0	67.8	66.0	62.4	55.2	51.9
	DBStd	7.83	4.46	4.53	4.90	5.16	5.25	5.23	4.45	4.69	4.68	4.94	4.91	4.57
	HDD50	141	49	35	10	4	0	0	0	0	0	0	10	33
	HDD65	2507	448	384	318	253	165	59	19	12	38	111	294	406
	CDD50	3478	65	71	159	208	324	449	526	552	480	384	167	93
	CDD65	369	0	0	2	7	24	58	80	99	68	30	1	0
	CDH74	2333	0	0	19	46	169	434	539	659	346	120	1	0
	CDH80	696	0	0	0	5	39	135	187	229	84	17	0	0
(14) Wind	WSAvg	6.9	7.2	7.3	7.6	7.4	7.3	6.8	6.4	6.1	5.9	6.7	6.9	7.3
(15) Precipitation	PrecAvg	49.3	6.5	5.8	3.9	4.5	3.7	1.9	0.6	0.9	2.6	5.2	6.5	7.2
	PrecMax	78.9	16.7	15.2	14.4	15.7	9.2	10.0	2.8	3.5	7.6	13.7	19.1	24.6
	PrecMin	33.9	0.2	0.7	0.0	0.4	0.1	0.0	0.0	0.0	0.0	0.5	0.6	1.1
	PrecStd	9.7	3.9	4.0	3.3	3.2	2.4	1.9	0.7	0.9	1.9	3.4	4.1	5.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	68.8	76.7	79.2	84.9	89.5	91.3	92.5	87.3	82.3	71.7	65.5
		MCWB	54.1	55.1	56.1	61.2	63.4	65.4	67.2	66.9	64.2	63.3	58.9	54.9
	2%	DB	60.4	63.7	71.2	73.7	78.8	84.0	85.3	86.2	82.2	77.2	67.6	62.5
		MCWB	54.2	53.0	54.5	58.4	61.9	64.0	66.3	65.8	63.4	61.8	58.4	55.6
	5%	DB	58.8	60.4	66.4	69.4	73.6	79.0	80.1	80.9	77.7	73.4	64.7	60.6
		MCWB	54.4	52.5	54.2	57.5	60.6	62.9	65.5	64.5	62.8	61.2	58.7	56.1
	10%	DB	57.4	58.3	62.6	65.6	69.3	74.3	75.1	76.6	74.0	69.7	62.7	59.0
		MCWB	53.8	52.7	52.9	56.2	59.4	62.6	64.7	64.1	62.6	61.1	57.6	55.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	58.6	59.8	63.9	67.0	69.5	70.6	71.4	68.4	67.9	64.6	60.9
		MCDB	60.2	61.7	67.2	73.5	77.5	80.4	82.9	81.4	77.1	72.7	65.9	61.3
	2%	WB	57.2	56.9	57.8	60.8	64.1	67.3	68.6	68.4	66.5	66.0	62.1	59.0
		MCDB	58.4	59.5	63.8	68.7	73.2	77.8	79.4	77.0	73.9	70.5	64.4	60.2
	5%	WB	56.0	55.5	56.6	59.0	62.3	65.5	67.0	67.1	65.4	64.6	60.5	57.4
		MCDB	57.5	58.1	61.6	65.9	70.4	74.1	75.8	74.8	72.2	69.0	63.1	59.4
	10%	WB	54.8	54.1	55.4	57.5	60.7	64.1	65.7	65.9	64.5	63.3	58.8	55.8
		MCDB	56.8	57.2	60.1	63.7	66.9	71.4	72.9	73.1	70.7	67.8	61.7	58.3
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	13.7	14.3	14.0	13.5	14.2	14.5	15.5	15.1	13.8	12.9	12.0
		MCDBR	12.7	16.5	21.3	20.5	20.6	22.6	23.0	23.2	21.1	19.3	14.6	12.4
		MCWBR	8.9	10.0	11.6	10.7	10.2	10.2	10.1	9.7	9.1	9.0	9.0	8.7
	5% WB	MCDBR	8.1	11.6	15.6	16.9	16.3	17.5	17.7	16.9	15.3	13.0	9.8	9.0
		MCWBR	7.6	9.0	9.9	9.9	9.0	9.0	9.0	8.7	8.2	7.7	8.0	8.2
(40) Clear-Sky Solar Irradiance	taub		0.332	0.342	0.375	0.374	0.382	0.385	0.375	0.387	0.376	0.368	0.340	0.328
	taud		2.503	2.465	2.380	2.391	2.395	2.402	2.420	2.384	2.437	2.475	2.509	2.510
	Ebn at Noon		260	275	277	285	284	282	284	277	273	262	255	252
	Edh at Noon		24	29	35	36	37	37	36	36	32	28	23	22
(44) All-Sky Solar Radiation	RadAvg		585	911	1257	1662	2026	2208	2250	2033	1584	1020	667	529
	RadStd		99	128	185	187	177	146	116	107	113	120	114	80

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.59	N/A	N/A	N/A	-0.56	N/A	N/A	N/A	+194	+72
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70

Nomenclature: See separate page